Assignment 5

September 6, 2026

Given:

- $m_{H^0} \approx 125$ GeV$/c^2$
- $m_{Z^0} \approx 91$ GeV$/c^2$
- $m_e = 5.11 \times 10^{-4}$ GeV$/c^2$
- $m_\gamma = 0$

1 Part 1

the maximum energy of the electron at LEP was

$$E = 104 \text{ GeV}$$

the total relativistic energy of a particle is

$$E = \gamma m c^2$$

rearranging:

$$\gamma = \frac{E}{mc^2}$$

rest mass of the electron is

$$m_e = 5.11 \times 10^{-4} \text{ GeV}/c^2$$

so, solving for γ

$$\gamma = \frac{104 \text{ GeV}}{5.11 \times 10^{-4} \text{ GeV}}$$

$$\gamma \approx 2.04 \times 10^5$$

therefore to 1% accuracy,

$$\boxed{\gamma \approx 2.04 \times 10^5}$$

(super close to the speed of light)

2 Part 2

we know that Lorentz factor γ is:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

squaring both sides,

$$\gamma^2 = \frac{1}{1 - \frac{v^2}{c^2}}$$

rearranging,

$$\frac{1}{\gamma^2} = 1 - \frac{v^2}{c^2}$$

factorising the RHS,

$$1 - \frac{v^2}{c^2} = \left(1 - \frac{v}{c}\right)\left(1 + \frac{v}{c}\right)$$

therefore,

$$\frac{1}{\gamma^2} = \left(1 - \frac{v}{c}\right)\left(1 + \frac{v}{c}\right)$$

for a highly relativistic electron, its velocity is very close to the speed of light, so

$$\frac{v}{c} \approx 1$$

therefore,

$$1 + \frac{v}{c} \approx 2$$

substituting this gives:

$$\frac{1}{\gamma^2} \approx 2\left(1 - \frac{v}{c}\right)$$

rearranging,

$$\boxed{1 - \frac{v}{c} \approx \frac{1}{2\gamma^2}}.$$

3 Part 3

the dominant reaction for H^0 production is

$$e^+ + e^- \to H^0 + Z^0.$$

electron and positron have equal and opposite momenta, so the centre-of-mass frame is the same as the laboratory frame. to find the minimum energy required, we consider the reaction at threshold. at threshold, the final particles have zero momentum in the centre-of-mass frame, so they are produced at rest relative to each other.
this means that the centre-of-mass energy is just enough to provide the rest energies of the H^0 and Z^0:

$$m_{H^0}c^2 + m_{Z^0}c^2$$

using

$$m_{H^0}c^2 = 125 \text{ GeV}$$

and

$$m_{Z^0}c^2 = 91 \text{ GeV}$$

we get

$$125 + 91 = 216 \text{ GeV}$$

since the electron and positron have equal energies,

$$2E_e = 216 \text{ GeV}$$

therefore,

$$E_e = \frac{216}{2} = 108 \text{ GeV}$$

hence, the minimum energy required for each electron and positron is

$$\boxed{E_{e,\min} = 108 \text{ GeV}}$$

4 Part 4

4.1 (a) $H^0 \to e^- + \bar{\nu}_e$

forbidden. charge conservation is violated. H^0 is electrically neutral, but in, $e^- + \bar{\nu}_e$, the electron is negatively charged and the anti-neutrino is neutral, making it have a net charge of -1.

4.2 (b) $H^0 \to Z^0 + Z^0$

forbidden. this fails on energetic grounds. the invariant mass of the final state must satisfy $m_{\text{final}} \leq m_{H^0}$ for the decay to be kinematically allowed. here,

$$2\, m_{Z^0} c^2 = 182 \text{ GeV},$$

which exceeds $m_{H^0} c^2 = 125$ GeV. there is not enough rest-mass energy available in the parent particle to produce two on-shell Z^0 bosons.

4.3 (c) $H^0 \to Z^0 + \gamma$

allowed. charge is conserved ($0 \to 0 + 0$) and the mass budget is satisfied (91 GeV $<$ 125 GeV).

4.4 (d) $H^0 \to Z^0$

forbidden. a single particle cannot decay into a single particle of a different mass. since $m_{Z^0} \neq m_{H^0}$ (91 GeV $\neq$ 125 GeV), energy and momentum cannot simultaneously be conserved, so this process is not possible

4.5 (e) $H^0 \to \gamma + \gamma$

allowed. charge conservation holds. photons are neutral and massless, so kinematics are easily satisfied (125 GeV $>$ 0).

5 Part 5

assuming that the decay $H^0 \to Z^0 + \gamma$ is possible.
the masses are

$$m_{H^0} = 125 \text{ GeV}/c^2$$

$$m_{Z^0} = 91 \text{ GeV}/c^2$$

$$m_\gamma = 0$$

5.1 (a) Q-value

the Q-value is given by

$$Q = \sum m_{\text{initial}} c^2 - \sum m_{\text{final}} c^2$$

there is one initial particle- the Higgs boson, and two final particles- the Z^0 and photon. therefore,

$$Q = m_{H^0}c^2 - m_{Z^0}c^2 - m_\gamma c^2$$

substituting in the given masses,

$$Q = 125 - 91 - 0$$

$$Q = 34 \text{ GeV}.$$

therefore,

$$\boxed{Q = 34.0 \text{ GeV}}.$$

5.2 (b) momentum of γ and Z^0

we consider the decay in the rest frame of the Higgs.
since the Higgs is initially at rest,

$$\vec{p}_{H^0} = \vec{0}$$

conservation of momentum gives

$$\vec{p}_{Z^0} + \vec{p}_\gamma = \vec{0}$$

the two particles have equal momentum and travel in opposite directions:

$$p_{Z^0} = p_\gamma = p$$

for a relativistic particle, the energy-momentum relation is

$$E^2 = p^2c^2 + m^2c^4$$

for the photon,

$$m_\gamma = 0$$

so

$$E_\gamma^2 = p_\gamma^2 c^2$$

taking square root on both sides,

$$E_\gamma = p_\gamma c$$

for the Z^0,

$$E_{Z^0}^2 = p_{Z^0}^2 c^2 + m_{Z^0}^2 c^4$$
$$E_{Z^0} = \sqrt{p_{Z^0}^2 c^2 + m_{Z^0}^2 c^4}$$

substituting into the energy conservation equation and using the relation defined for p_γ and p_{Z_0}:

$$pc + \sqrt{p^2c^2 + m_{Z^0}^2 c^4} = 125$$

rearranging terms,

$$\sqrt{p^2c^2 + m_{Z^0}^2 c^4} = 125 - pc$$

squaring both sides:

$$p^2c^2 + m_{Z^0}^2 c^4 = 125^2 - 2(125)pc + p^2c^2$$

the p^2c^2 terms cancel on both sides

$$m_{Z^0}^2 c^4 = 15625 - 250\,pc.$$

substituting $m_{Z^0}c^2 = 91$ GeV:

$$91^2 = 15625 - 250\,pc$$

$$8281 = 15625 - 250\,pc$$

$$250\,pc = 15625 - 8281 = 7344$$

$$pc = \frac{7344}{250} = 29.376 \text{ GeV}$$

$$\boxed{p_\gamma = p_{Z^0} \approx 29.4 \text{ GeV}/c}$$

6 Part 6

the decay in question is

$$Z^0 \to \mu^- + \mu^+$$

the Z^0 boson is a neutral gauge boson associated with the weak interaction. therefore, the fundamental interaction responsible for this process is the

$$\boxed{\text{weak interaction}}.$$